



PIXEL

vienna
Review

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**Ken Fountain's
Animation Workflow**

Interview with Blizzard Talent
David Luong

**The Concept Art of Ori &
The Blind Forest**
Interview with Johannes Figlhuber

AUSTRIAN COMPUTER GRAPHICS & ANIMATION CONFERENCE MAGAZINE

SUPPORTERS 2015

IG-COMPUTERGRAFIK
ANIMATION, COMPUTERFILM UND COMPUTERSPIELE



TELEGRAM71
building visuals



arX anima
we animate



**OESTERREICHISCHE
COMPUTER GESELLSCHAFT**
AUSTRIAN
COMPUTER SOCIETY



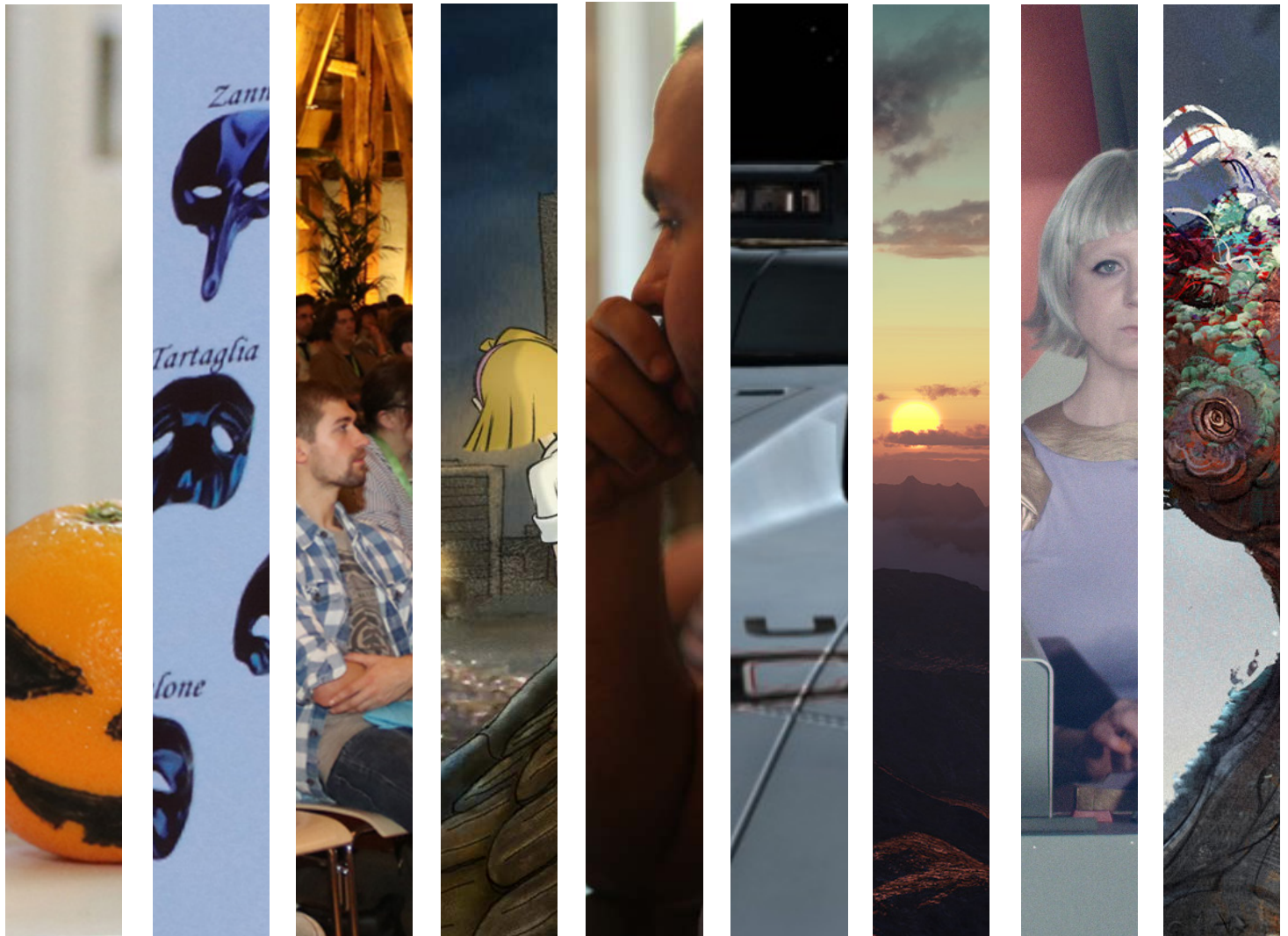
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PIXELis...

Online Forum & Magazine
www.cgmag.at

Austrian Computer
 Graphics Award
www.acga.at



Lobby for
 Animation,
 Digital Film
 and Games

www.igcomputergrafik.at

10th Anniversary



PIXEL 1

March 24th, 2007
1 day conference
90+ visitors
10 talks

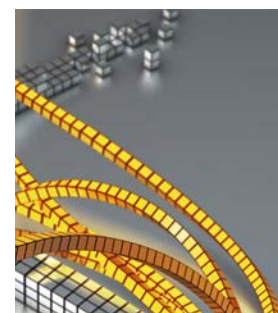
Christopher Evans - Crytek
Hanne Hegedorn - Crytek
Wolfram Neuer - RASCAT
Ralfmann Schmitt
Christian Schmitt
Christian Staber
Mario Reibauer
Thomas Wabner
Ronald Kalchauer - Games College
Reinhold A. Proger - Industrial Motion AG
Christian Freyberger
Anna Czerwik
Tobias Hahn




PIXEL 2

Sept. 8th - 9th, 2007
1st time 2 days conference
100+ visitors
13 talks
1st Pixel Party & Battle

Finzer MacLean - Disney
Rüdiger Rask - Mental Images
Johannes Muecke - Spring
Stefan Kubisch - Spring
Michael Muehling
Christoph Staber - RASCAT
Stefan Auer - Wunderwerk
Roman Barzila - FriendlyFire
Christian Stenjar - Basic
Reinhold Schmid
Reinhold A. Proger - Industrial Motion AG
Christoph Schmitt
Ronald Kalchauer - Games College
Mario Reibauer

PIXEL 3

Oct. 25th - 26th, 2008
120+ visitors
10 talks
1st speaker from overseas

Leo Stringer - Lucastim Animation
Alvaro Gascón - Orange Film
Hagen Krey - Ambient Entertainment
Florian Sattlinger
Marcum Laura Mazzoni - Sattlinger & Mandenberg
M. Wernel, M. Blum, B. Griesner - Comedy
Johannes Muecke
Reinhold Schmid - RASCAT
Christoph Schmitt
Martin Pusch - RASCAT





PIXEL 4

Sept. 25th - 27th, 2009

New conference location

1st film festival

250+ visitors

16 talks

Michael Seitzmann
Daniel Heck, Helen Antonen - Blue Zoo
Tobias Hannen
Julian Kneeling - Bear Silver Vienna
Gowth Kozani - Team Vienna
Wayne Robson - Redbox
Hector Moran
Alvaro Galindo
Ronald Anzenberger - Kasia Online
Florian Sattlinger
Marcel Louis Haezdi
Toni Weiss
Christoph Quast - Spraying
Reinhold Prager - Industrial Motion Art
Reinhold Prager, Michael Langmayr - Autodesk



PIXEL 5

Nov. 5th - 7th, 2010

250+ visitors

17 talks

Sebastian Weisold - Pixar Animation Studios
Harald Siegmann
Jakob Schuch - Studio Set
Johannes Mücke - Wideshot
Stefan Luch - LAUBlab
Ryan Listott - Squint/Opera
Patrick J Palmer - Sides
Joshua Beveridge - Sony Imageworks/Aardman
Jules Tervoort, Micha Rozema - Post Panic
D. Hüttner, P. Zeymer, B. Kalchauer - Mi'pami Games
Günter Mikolaj - Cybertime
Hector Moran - Spraying
Mark Tramon
Sven Havemann - C&V, TU Graz
Arnold Hess - Eutrochem
C. Kragler, M. Stöger - GLN
Roman Barakat - FriendlyFire
Reinhold Prager - Industrial Motion Art
Lor Ladi - GAME DESTALT



PIXEL 6

Oct. 7th - 9th, 2011

300+ visitors

22 talks

1st educational track

Andreas Dajc - Disney Animation
Simon Teun - Blizzard Games
Christian De Vita - One Hand Clapping
Martin Aullinger - Fromstarters London
E. Kneist, T. Buchta - Bohemia Interactive
Eric de Broche des Combes - Lusion
Carl Calloway - Galley
Michael Nishitaky - PZVR
Matthias P. Richter - tickets8
Emmanuel Turquin - The Bakery
Eric Meitz - Postcard
Helge Moss - Pixeltrain
Cecilia Denk - RTT
R. Barakat, P. Zeymer, V. Strubel - FriendlyFire
J. Tamsack, J. Strass, P. Wiedig - AAA Studio
Coco Klostner - Side PE Software, Guerrilla Games
Heuer Österreichischer Trickfilm
Jörgen Hagler - FM Hagenberg
Orhan Kizilic - FM Hagenberg
Alexander Hofmann - FM Technikum Vienna
Filmakademie Baden-Württemberg
Institute of Computer Graphics and Algorithms - TU Wien
University of Applied Sciences Hagenberg
SAB Institute Vienna
Fachhochschule Salzburg University of Applied Sciences



PIXEL 7

Oct. 5th - 7th, 2012

350+ visitors

17 talks

1st Austrian CG Award

Party and Battle @ Rote Bar

Bruno Bozetto - Studio Bozetto & Co
Oscar Grillo - The Human Race
Michael Sheatle - Pixar Animation Studios
David Brooks - Aardman
Craig Caldwell - University of Utah
Thiago De Costa - TeamUp Technologies
Damien Fagnon - MPC
Fabio Taccu - Farge Studios
Michael Paack - Cliffhanger Productions
Oliver Reischl - Bongfish
Gerald Ziegler - RVEDA UK
Jarek Kalar - 3K Czech
Jochen Kraxner - ewe
Ben Long - WolzeBuffet
Roy Malisoff
Paha Schatz - Crytek
Toni Weiss
Donau Universität Krems
SAB Institute Vienna
Filmakademie Wien
FH Technikum Vienna, University of Applied Sciences
FH Salzburg, University of Applied Sciences
Joanneum, University of Applied Sciences
Institute of Computer Graphics and Algorithms, TU Wien
FH Hagenberg, University of Applied Sciences



PIXEL 8

Oct. 25th - 27th, 2013

3 Oscar winners on stage

350+ visitors

12 talks

Eric Goldberg - Disney Animation Studios
Trey Sallins - Sony Pictures Imageworks
Christophe Mery - Pixar Animation Studios
Markus Korte - Rhythm & Hues Studios
Mario Wolf - Pixelon-Pictures
Robert Skalsky - Praxis Film Works
Hannes Seifert - So-Interactive
Simone Kesterton - Beadall Entertainment
Martin Pichler, Clemens Scott - Broken Rules
Christopher Schmitt - Ubisoft Blue Byte
Maciej Szczepanik - CB Projekt RED
Roman Wladis - 3K Czech
Petr Babek - Ateliery Zlin
Michael Hengst - Flying Fishkopp Productions
Hannes Seifert - 3D Interactive
Johanna Schaber - Spraying
Michael Paack - Cliffhanger Productions
Ralph Nebel - Disney Research Zurich



PIXEL 9

Nov. 7th - 9th, 2014

400+ visitors

17 talks and workshops

Jose Lopez - Beasler Inc.
Andrei Babitsch
Alex Ortel - Crew 872
Eric Tenner - Stop! Buddy Studios
Celine Graham - Ubisoft Montreal
Leo Stringer
Alvaro Galindo
Stefan Auer - Wunderwerk Film
Joao Cabada - Next Limit Technologies
Andreas Mennel, Filippa Viviano - Spraying
J. Biedler, M. Goltmann - Cliffhanger Productions
Vassil Papayanov - Chaos Group
Andy Touch - Unity Technologies

A man, Ken Fountain, is standing in front of a large projection screen. He is wearing a dark t-shirt and glasses, and is pointing his right hand towards the screen. The screen displays a colorful animation of several gnomes. One gnome on the left is holding a green octagonal sign that says "GO". The gnomes are wearing various colored hats and robes. The background of the projection is a soft, out-of-focus landscape.

KEN FOUNTAIN'S ANIMATION WORKFLOW

Interview by Michaela Jindra

Ken Fountain is a feature film animator, an instructor at iAnimate and also the founder of both SplatFrog and Crackerbox Studio. Films he has worked on include "Megamind", "Kung Fu Panda 2", 'Puss in Boots' and most recently 'The Peanuts Movie' ...to just name a few. CGmag had the pleasure of an interview with him after his talk at PIXELvienna X.

What does your animation workflow look like when you get assigned to a new shot? Do you do a lot of planning like thumbnailing or shooting video reference, or does it mostly come directly out of your head? Did your process perhaps change over the years?

Yeah, it has changed. But, I do have a pretty standard way of working, at least when I am given enough time, because it is different when you have a really fast job, of course. At first, you need to ask a lot of questions. I think that is something that animators are afraid to do a lot, because you are working with a client or you are working with a director or just your supervisor and they are your superiors, so you are intimidated. But, you have to really ask a lot of questions, because our job as animators is to interpret what is inside of their head, not what is inside of our head. So that is the first thing. I try to analyze with little drawings first, usually. I write down notes on what I think the character is thinking at different moments of the dialogue. I like to get quick sketches out, so I know how I am going to stage the shot and how I want to

fill the frame at different points. I do that before I get in front of the camera for video reference, so that I am not doing just random things in front of the camera. I am not big on doing like a hundred takes of anything because, like you said, sometimes it is more about what is in your head. I think video reference is only me "taking notes" and I know a lot of animators who think like that too. It is just a note - taking device and then you just pull out pieces of it that come close to what you see in your head as the vision - and then you are supposed to take that further. Those who went to my talk may remember... I am an evangelist when it comes to separating people from their video reference: References are your notes and then you need to do something better than that. So that is kind of my common thing. It is really asking questions, doing some little sketches, taking enough video reference to have the information that I need and then beginning to animate... beginning to block. And again, during that process, all I am really looking to find is the story from those poses. I am not necessarily looking for mechanics and all that yet. I want to make sure that my story poses, the ones that I want my audience

" ... ask a lot of
questions ... "

to remember, really are the best that they can be. So I spend way more time on that, than I do really fussing with movement. Which does not really make sense because animation is movement, but animation is a language, so those poses have to really be memorable.

So then for you, the actual physics just really come into play when you are already working inside of Maya?

Yeah, my video reference helps me analyze some of the physics, but we are exaggerating that. So my physical breakdowns come from my imagination more often than they do from real life. Unless I am doing something like dancing, a very specific choreographed dance or choreographed fighting, where there has to be like real karate... that is different of course. But just regular acting, the physical part, the mechanics usually come from my imagination more than from the video.

Was there ever a character or some particular shot that you were working on, that you really had difficulties with to get right? If so, how do you handle such situations and manage to get past them?

When I talk to my students, I think they are always happy to hear how much of a struggle animation is for everyone. Which is funny to say, but at some point in every performance you are creating, you get to that point where you are sitting there going "Aaaaah, I just cannot get it right!". You are frustrated and you are stuck on something. And that just means that it is challenging. If your shot is easy then it is not challenging, however, we always like the challenging shots. So, I am trying to think of what was really difficult... a specifically difficult character. I remember working with fat Puss in Boots was difficult, because his arms and legs did not really reach around his fat body. There were a lot of things you had to do to the rig, to deform it in certain ways, to get it to work right. Fat characters are really hard, especially CG fat characters, because the geometry is just difficult when things penetrate and when you wanna get that extra feeling of weight. And then, I specifically had a really hard time doing pretty much all of the shots I worked on in Kung Fu Panda, because I do not know martial arts

very well and, thankfully, there were people on the team who were martial artists and they could really, really help. And in fact, a lot of people on Kung Fu Panda did not do their own video reference. There will be those four or five people that knew martial arts. They could do those flips and punches and kicks... so that was difficult. I think that I am not very good at accurate, athletic, physical things. I can do it, but it is not what I think my strength is. I think I have more strength in, like, comedy and cartoony stuff. But I guess that all has to grow out of regular physics, but maybe I just do not like making things look real - I like making them look better than they really are.

Can you tell us a bit about your personal projects? What motivated you to leave the world of animated features behind - at least for a while - and focus more on SplatFrog and Crackerbox Studio?

For SplatFrog I just like teaching and I think I found something that I am good at, like explaining things. I feel like I have had success with that. So I wanted to do that, but I have always done that on the side. That did not really prompt me to leave feature film when I did. I mean, I have been back in and out of feature work, but I do not want to be permanently in it, because a project like Crackerbox is also educational, but it is really not like I am on screen teaching something. It is taking what I know how to do with character animation, storytelling, music and all that kind of stuff, explaining things and putting it all in one place, just to feel good about it. I grew up on these cartoons on Saturday mornings... they were little interstitials and they were called "Schoolhouse Rock". I do not know how many languages they have been translated to, but they were huge when I was a kid in the States, and they taught you grammar, things about science, things about the government and those things stuck in my head for the rest of my life.

Every single thing that I learned from a little Schoolhouse Rock cartoon, I still remember and I still use. So I want to be a part of that tradition and do something that is not just showing up on McDonald's boxes, but something that little kids or even adults remember. And, I had some friends that were interested in doing such a project too. That helped. So, I knew that when



"... poses have to really be memorable ..."

Animating Eyes
"The Avocado and the Eyeball"
SplatFrogAnimTip #03

(<https://youtu.be/53rqwGFAj4o>)



my contract was up, I could call my friends and we could give it a try. It is a slow process. We have not produced a ton of things so far, but it is something that we are still constantly working on. And that really makes me pretty proud. I mean, anybody would be proud of something that they get to do, which is just theirs. And I just want mine to be something educational, more than just entertainment.

So are you working on new apps right now or will you do more content for the Geo-Me app that is already out?

We will see. Geo-Me came out like two years ago and it was not getting a lot of traction. It was getting great reviews, but it was not selling very well. So, we could not afford to make more animation for it. And then, like a month ago, this educational company wanted to add us to their collection of educational apps that they have curated for teachers. And all of a sudden, now we are having sales again. In fact, in the last two weeks, we have had as much sales as we have had in the last year and a half. It was a random accidental thing, but what that is probably going to do, is make us go "Huh... ok, I guess I should continue to add some more animation to that!". But the project that I have started working on last year, and I am actually taking this month off to do some more work on, is the "Wizards In The Machine" one, which I am having a lot of fun with. Computers have just been something I have always understood very well, but it is something that not everybody understands. So, I want to create little scenes, that help demystify what is going on inside of a computer and the wizards just felt like a fun little thing. That was great, because it was creating everything from scratch... That is been fun and people liked it. I am working on that this month and I am trying to put together a few more

" ... just
start noticing
things within
other things ... "

storyboards, so I can either start out myself, or I have a couple of places I am going to pitch it to, to see if they want to pick it up. So we will see.

Because you mentioned teaching before: You have a talent of making pretty complex topics very easy to understand. For instance, I am thinking about the avocado and carrot video that you did about animating the eyes. Just awesome! So, how did you get into teaching in the first place? Was it something that you always had a passion for or was it more of a lucky coincidence that made you first start doing it?

That is the fun thing of what we do. All of a sudden, you just start noticing things within other things. That is just the way I think. And now, when I walk around, my whole life is just thinking about relating things to animation - which I guess, we all do. But I did not really pursue going into teaching. It all started when I was working at Dreamworks.

My friend Jason Ryan, who was well known for his tutorial videos, was doing some online seminars and then he also began to tutor classes of people and it was all getting a little overwhelming for him. So he kept asking his friends to also give this a try, so that people would have more choices. But we were like "Meeeh, I do not want to do this. It is too much work and I just want to animate" and then, he went "Well screw it, I am going to start a school". And so he got a business partner and put together iAnimate and then of course, asked all of his friends to come and be teachers and contribute. Of course then, when we did not have to do any work inbetween projects, we were like "Oh yeah, that would be great!". So, I just gave it a try and within the first couple of weeks of teaching I was like "Wow, this is awesome. I can do



this!". And then Jason again... I feel like he encouraged so much of it, because he suggested that I should start making some videos too. Like the videos that he did, so people would have more choices. So, when I had some time, I gave it a try and made my first one. It was successful, so I made a bunch more. You see, it was really kind of just an accident and I decided that I liked it. And I want to keep doing it! I am finding tons of different ways this year to travel and to do presentations at conferences and at people's studios. I am also going to be working with some art colleges in the States now - like coming in as a video professor, kind of. So that is great and I am having a blast.

You recently joined the crew at EVIL EYE PICTURES who are doing one of the next animated shorts for Google Spotlight Stories ("Pearl", directed by Patrick Osborne). Can you tell us a little bit about that project? Or is it still top secret?

I was working on that project from the middle of August through the middle of November. It was really fun! The fun thing about it is, that you... well, I actually do not know if I would describe this part as fun, because the specific challenge in this medium is, that you are not just working in the frame. And I have never worked in games. So maybe it is easier for people who have worked in games, but for a traditional movie or series, when I set up a shot, it is all about the composition and the movement in the frame and even if it is a moving camera, there are still limits. This means, you are tempted to get lazy, because you do not have to animate anything that is not in the frame. But in these interactive 360 degrees stories, you can look anywhere you want. So there is extra work, because you have to make sure everything is animated, but you also have a bigger challenge of focus. And this is something that I teach a lot in my advanced classes, when we are working with cameras and stuff: Keeping the audience looking where they need to look. Well, now imagine, if they can move the camera wherever they want. You have to create the action in such a way that they do not want to move the camera or look somewhere else.

So, between the talent of the director, who really thought long and hard about how to move focus in this infinite space... he did such a great job at keeping us focussed. We were all seasoned professionals, so we were able to kind of figure out what he wanted fast and it was really fun. It was a very small team. There were only eight animators on the whole team and then, by the end of it, there were only four of us that stayed on for the long haul. I have never contributed so many shots to one project in my life. It was great. It is a really touching little story. It is about a father and his daughter and it's got lots of music and the art direction is just beautiful. I



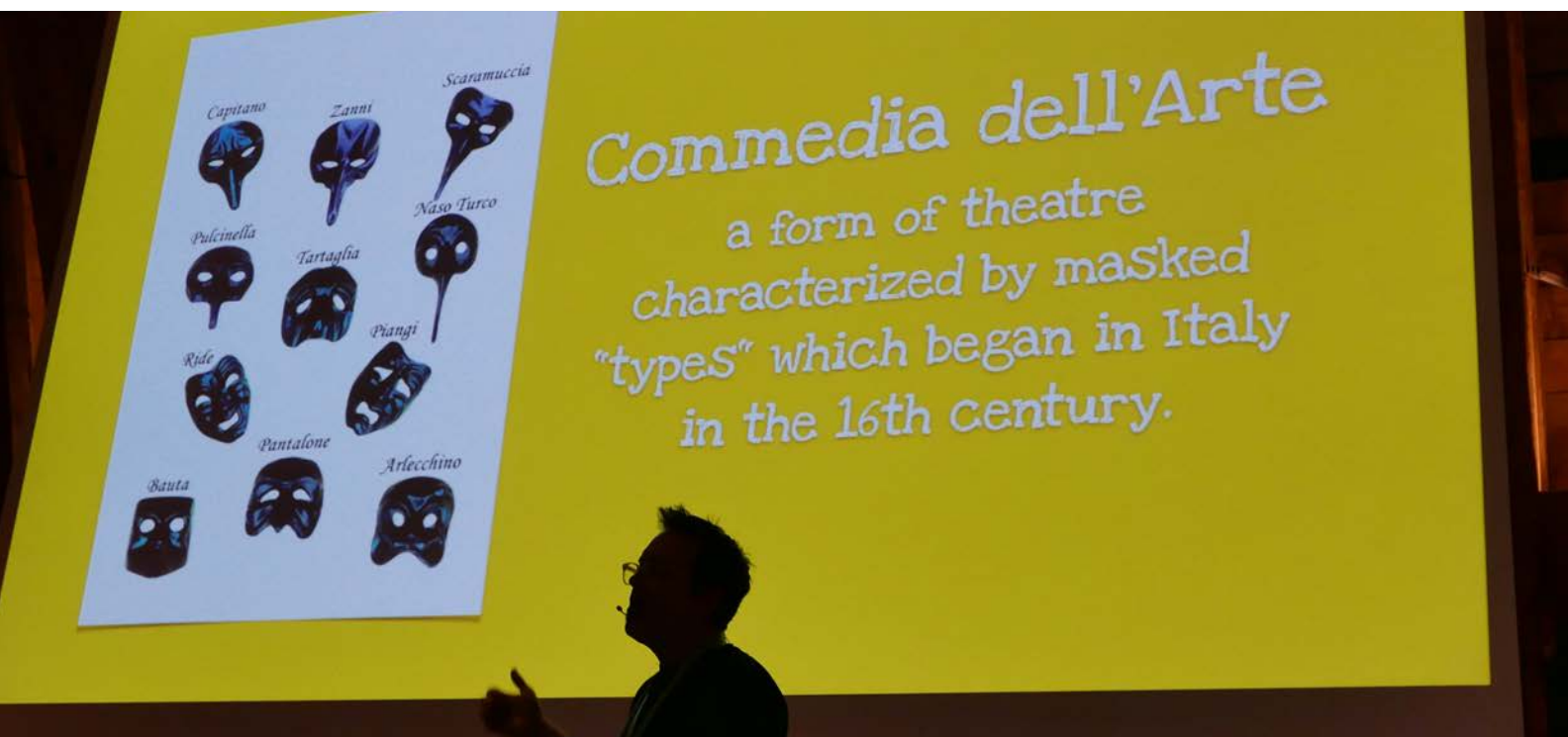
have not seen much of the final product, so I am just trusting it is going to be great. Because the people that are working on it are really talented. And it was online too... it was remote work for everybody.

How does storytelling work in 360 degrees? You already talked about some of the limitations compared to traditional movie making, but how can it, on the other hand, inspire completely new and creative ways of doing things and telling stories?

I hope that it does create new ways of telling stories, because I really like the virtual reality theatre that is now coming to life. I worked in regular theatre for a long time in my late teens and early adulthood and I describe this to my students a lot too. In theatre you have this big stage and your audience is sitting there. They have the ability to look anywhere they want. It is the director his job and your job as a team of actors to make sure that the specific point on that stage is being looked at at a specific time. It is different from film, because in film you can just point the camera.

In theatre you are competing with the guy sneezing in the back row or somebody's cellphone going on or an exit sign that is flickering over the door. There is so many things that you have to be really good at focusing. I love that now it is sort of breaking out of that static camera and we are challenged again to really work harder to entertain an audience in a specific way. So they look at a specific place and follow a specific story. And the other side of that is, you can have, like Glen Keane, intertwining stories that happen and so you can choose to follow another one the next time you watch it. All that, just as an animator, I love that. I want that challenge. I think it is really cool.

"... Keeping the audience looking where they need to look ..."



With Google Spotlight Stories and also the Oculus Story Studio... Do you think that might be the future of storytelling and how we watch movies? Like really immersing the viewer into the action and giving him some sort of power over the whole thing?

It is going to become a popular way of telling stories. Yeah, but I think there are just going to be more choices. I do not think theatre is going to go away too soon, because that is an art form. It is like saying painting has gone away, because computers happened. There is always a place for the beauty of a single frame.

My last question is one that you have probably been asked like a hundred times before. But I am still going to ask it: What advice would you give young artists, who want to get into the animation industry?

Well, I have a couple of answers for that. And I have been thinking about this a lot, especially now that I am starting to do some work with art schools... I never went to art school, so I am just starting to understand how it really works... The animation industry is a lot of different things. And you may discover that you like one specific area of it, or you may discover that you like all of it. And either way, there is a place for you. Because smaller studios require you to have a vast amount of skills and larger studios require you to have a really specific kind of skill. But when it comes to just animating, people learn really fast that it is not easy and I think it is a reality.

If you are not the kind of person that can obsess over one thing until you forget to eat and forget to sleep... if you are not that person, you may not like animating. It takes a lot of work. It is

like saying, I want to play guitar and expecting to know how to play in like 18 weeks or whatever. It is going to take you years and years and hours of hours of just doing it and then, all of a sudden, you will wake up and you will go "Oh my god, I know how to do this!". But it takes a lot of obsession. So, that sounds terrible and I do not like to discourage people, like "Oh, you have got to be strong enough", but it is ok if you are not, because there is tons of other really cool things inside of the animation industry that people have discovered or that they like. And so, I think that is why people... You know, when you start off with a class of 30 people, you probably get two, that will actually ever really become animators, because it does take a lot of work! But still, it is fun. I guess, the positive advice I would give is that all of the silly, for instance, how you just watch things the same way I watch things and I record them and kind of analyze them in the back of my mind.

There are a lot of people doing that, just looking at stuff and making comparisons and who want to learn how things work. You know, all of that stuff that you thought you would never have a use for, animation is where it gets used. So, I think it is a really great thing for people to discover. Like "I just thought it was weird and I will never be able to use this kind of thing in my brain" and then it is like "Yeah, you can. This is right where it belongs". Even if you are like a musician or you are a dancer or whatever. I worked with a kid, who was a car mechanic. He worked at a body shop his whole young adult life and it was his first animation job. He would constantly make comparisons to body work on cars and I will be like "That is why you are an animator. It is kicking ass that you can make connections with everything in life!". You just need to open up and then there is a place for you.

"... animation industry is a lot of different things ..."

PREVIZ THE FUTURE?!

VISUAL EFFECTS & VIRTUAL REALITY

Article by Axel Dietrich



ALBERT HASTINGS

CG-Supervisor @ Quantic Dream

Like so many of us in the digital film industry, Albert started his career after being inspired by the Star Wars films and the impressive universe created around them. Albert started out as a cameraman and later worked as VFX artist in Australia. He realized soon that as a VFX artist, one was only able to concentrate and develop a very small, highly specific section of a big project. He found this little rewarding and started looking for other aspects of film projects in which he could bring more factors into play. This is how he started looking into previsualization, where he could influence camera, timing and animation, thus working on the movie as a whole. "You basically make the movie in previz" Albert explained.

Since the 90ties, previsualization has become increasingly more sophisticated and a key part of animated movies and several other film projects in which VFX are required. Albert referred to technologies such as virtual camera, motion capturing and next gen game engines.

A virtual camera is a physical rig that is operated like a normal camera rig, but with the main advantage of having all movements of the camera read out in real time and sent to a digital camera in a constructed digital environment. This allows a cameraman to operate a digital camera in a very intuitive way and see the results in real time.

Motion Capture allows the recording of a character's movement to a level of detail that is almost impossible for an animator to achieve. Even the recording of facial expressions are no longer a big deal.

Due to the emerging of more precise and affordable solutions in this area, this technology is becoming a standard tool for previz. Next gen game engines and the development of GPUs in the last years have made it possible to get images in real time that are very close to the final movie.

This technology, combined with virtual cameras and motion capturing, now makes it possible to film CG heavy shots in real time, review them on set and send detailed shots to the VFX department. There, they can reuse most of the already created elements for the finishing process,

rather than creating everything from scratch. As a previz artist, nowadays, you are a very close companion of the director and the film crew on set. Both the director and the crew are in constant need of being able to try out different things quickly. The previz artist facilitates fast solutions for them.

Albert also talked about his future thoughts. According to him, VR is going to change the film industry significantly over the next two to three years. New affordable and technically advanced HMDs (head mounted displays) such as the Oculus Rift and new technologies are the main drivers of this development.

He showed us a presentation video of the ILMxLab and their vision of what the future of Virtual Reality will be on Youtube. ILMxLab is a collaboration of ILM, Skywalker Sound and Lukasfilm, with John Gaeta as creative director.

In this presentation we watched a scene from Star Wars in which you could move around freely to relive the same scene an infinite amount of times from different perspectives, much like

in a computer game, but with impressively realistic graphics. In different parts we could see a scene with a velociraptor, that was animated by a puppeteer in real-time, while the viewer could interact with it from a different part in the studio.

All the techniques are still in development and it is not clear how stories will be told in VR in the future. Virtual reality is in a similar situation like the medium film was in its early days. In over a hundred years, the language of this medium has developed and expanded a lot.

Now we have to discover how VR works. There are a lot of questions to be answered concerning VR. But one thing is for sure: No matter if you are a director, a camera man, a VFX artist or a game developer, there will be a place for you in Virtual Reality productions.



Cover-image of the game
"Ori & The Blind Forest"

THE CONCEPT ART OF ORI & THE BLIND FOREST

JOHANNES FIGLHUBER
Austrian Concept Artist

AND THE BLIND FOREST

Interview by Florentin Rieger

You studied MultiMediaArt at FH Salzburg. How did you start your further professional career?

I already knew quite early on that I wanted to do something in the illustration or concept design fields. I started with a five year HTL school for graphic design in Linz and got my relevant basic education there. I have always liked to draw, but did not dare to do something with that professionally right after school. So I applied the Bachelor in MultiMediaArt at FH Salzburg. Also because it seemed useful to gain knowledge in 3D and so on, if I plan to work in the industry. And it turned out to be a good decision. I learned 3D and worked a lot with Maya and 3D Animation, which I still benefit of. I can talk with the 3D artists at their eye level in their own jargon, I can speak with them about technical details.

After the Bachelor studies, I did an internship as a concept artist in a games studio in Oslo, Norway. I liked it and I wanted to stay there, but then the economic crisis came... They dropped a lot of employees and I started working as concept artist. Then, I worked for one and a half years for a company in Slovakia and after that, finally, I dared to become self-employed. I moved to Berlin and joined a community office of concept artists. I am still working there and we now call ourselves Airborn Studios.

On the internet one can find a project of yours called Airborn. Is that the origin of the studio name?

Yes, indeed. It started in 2005 as a hobby project and we kept on working on it. We liked the setting and style and I did some pictures for my portfolio. For some time, we really thought of making a game out of it, but we did not have any game designers, producers, animators or programmers in our team, so it was sure to fail. We have never dared to take the plunge, because it would be a very large project if we implemented it exactly as we imagined. This is somehow a pity, but let us see, maybe the time will come.

That means Airborn Studios is a group of friends?

Yes. We got to know each other over the internet. Most of us are German, some are American and there were two Austrians. And it developed. Some of us lived and worked in Berlin that is why I also moved there. As a self-employed person, I could have worked everywhere, but Berlin is a beautiful city.

For Ori And The Blind Forest, you worked in a decentralized team. What are the pros and cons for you?



Presentation of Johannes Figlhuber



Artist-Talk with Johannes Figlhuber

“ ... it is very important that there is still passion ... ”

Advantages: You have not to put on pants in the morning, definitely an advantage. Otherwise, you have all the benefits and all of the disadvantages of self-employment. You have to take care of everything on your own. You have to organize your work hours, take care of your office, your work computer, your software and taxes. These are pros and cons. Personally, I like to be responsible for myself. Other people dislike it, they prefer the false security of an office job. A definitive disadvantage might be the lack of contact with other people, but that was the reason why I moved to Berlin and joined the co-working community office.

How do you fight creative blockages, when you have the feeling of not moving ahead with your work?

This happens surprisingly often, it is something completely normal. You have your ups and downs. Ideally, people take their time out, taking a walk, making something else, reading a book, but nothing creative. Of course, that is not always possible. Sometimes there is pressure. There are some tools and methods to deliver efficiently solid, but not super creative work. There are rules and a lot of specialized literature on the topic. These are processes, you do not always like the outcome, but that is also a part of the deal in the creative industry. You get paid for functioning, that is sometimes frustrating, but most of the time it works.

Can you spontaneously name a book or a blog that deals with such tools and methods?

A very good example is Muddy Colors. There are repeatedly stories about that topic. It is always a big issue: creativity and frustration. Do I use my skills the right way? It is a permanent problem for everyone that works in a creative field.

Since you have made your passion into a career, do you still draw in your free time?

Lately, I have been a bit lazy, but I still have sketchbooks in which I draw when I am just bored or if I have an idea. I do not work so much digitally or much at home, because I am usually already exhausted after my work and then I do not want to start a new private project. I like drawing for relaxation. It is still my passion, despite it also being work now. I think it is very important that there is still passion, otherwise you will soon start to burn out. You have to find a way to make time for personal projects and find something beautiful in it. I hear from more and more artists that feel burned out and have achieved all their technical goals and do not know what to do next with their talent. This also happens to people that are among the



best artists in the world, these problems can hit everyone. We have to obtain meaningfulness and search for sense in our work.

How was that with Ori And The Blind Forest for you?

Ori was completely different. It was a project I loved, and admittedly, I am not sure if I would have been able to finish it, if I had not liked it, because it was really strenuous, as it is often with small game productions. There are difficulties and problems; you have to rework a lot and the finishing takes much longer than you would have thought. We had a planned buffer period of over a third of the time, but still did not finish it in time. Such times are very exhausting and you do not have any spare time for a private life. If the people like the project afterwards, if they give you the feeling it is something special, that is wonderfully rewarding.

A friend of mine also studied graphic design and wants to work as concept artist. What would you recommend him for his career in the games industry?

I would ask him: Where exactly in the games industry? Which, studios, games and styles do particularly interest him? I think it is better for beginners to focus on one project and a specific style. You should study the style of the studio you want to work for. You should at least pretend to know what you want. When you land your first job, you will have 40 – 60 hours working time to learn and improve – you never have so much time as a student.



them, and in return, they were inspired by our tools and they have developed a good toolset out of that.

Which software did you use for Ori And The Blind Forest?

For the communication, it was Skype and Google Hangouts. For some time, we also used Slack, but we stopped that, because the commercial version is relatively expensive and the free version was restricted. For documents and videos we used Dropbox and Google Drive. For versioning we used SVN (Apache Subversion) with Unity and for bugtracking and planning Hansoft. Animations were created in 3Dmax and textures in Photoshop. Unity was our game engine, I especially liked the workflow between Photoshop and Unity.

Is there another game you would like to work on?

I am a fan of role-playing games, for example, the new Fallout or Witcher 3. I am a fan of these worlds, to build up and explore which stories are behind the gameplay and how it is visually developed. I would love to work on such a project, that is fun. I like to develop worlds visually, that was the pleasure of working on Ori And The Blind Forest. It was a small team, so the responsibility for every person was high, which is more fun than doing just a small part in a huge project. I search for small projects, where you have a lot of freedom and self-definition as I find this to be more interesting.

“ ... focus on one project and a specific style ... ”

How did the co-operation with Moon Studios start?

I have known Thomas Mahler since long before. He is also Austrian and was very active on **cgforum.at** (Now: **www.CGmag.at**) and we had some contact. Thomas quit his job at Blizzard as cinematic designer, because he wanted to produce his own game. He pitched several ideas and needed visual support and I helped him non-profit with that. We worked together and over the years it became a close work relationship.

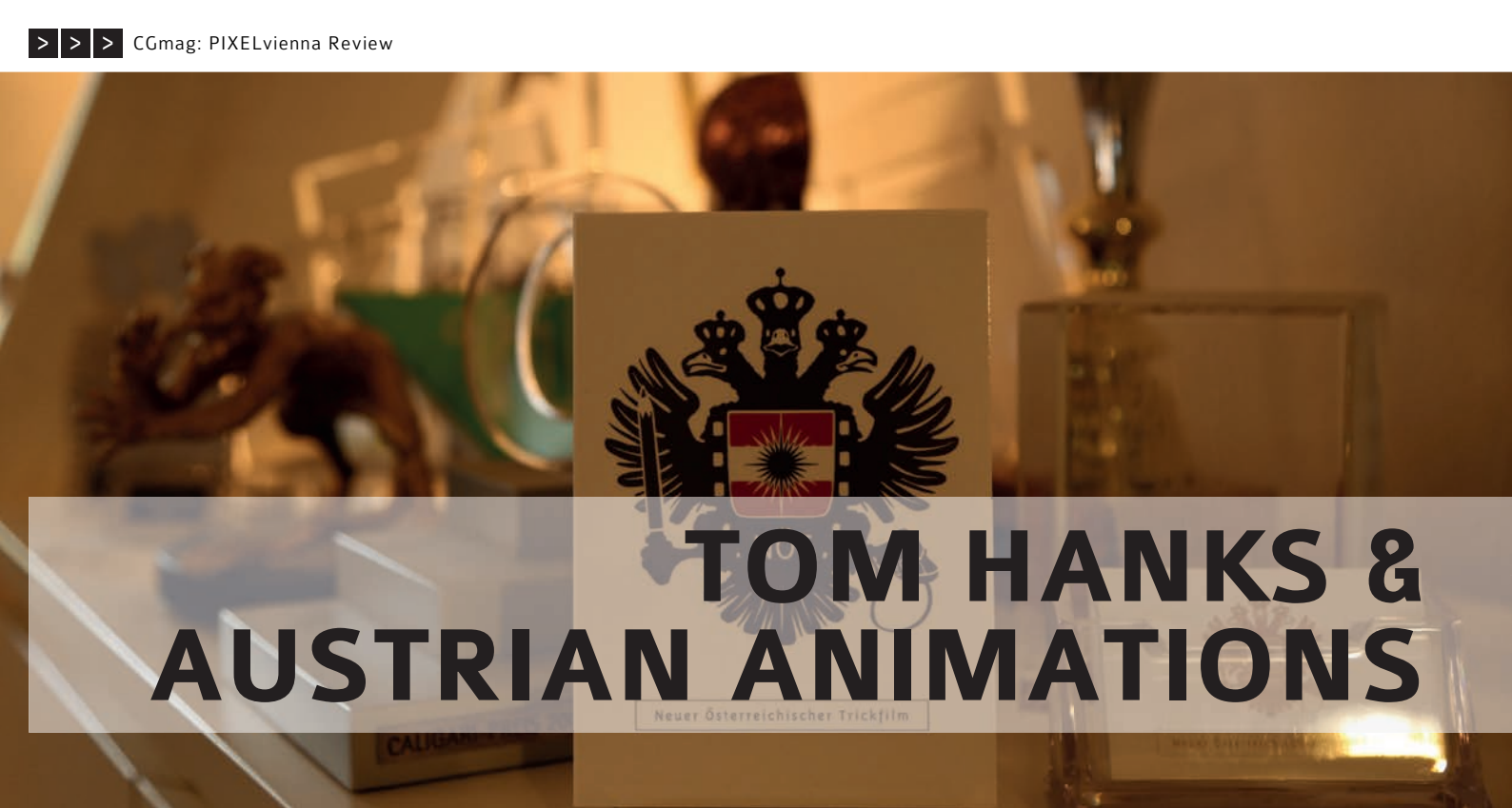
Did Thomas draw also some concepts for Ori And The Blind Forest?

Not really concepts: He did a lot of designs, he wrote and managed a lot. He was more the game director. He gave a lot of feedback, he had the vision and I tried to express it as lead artist.

Ori was developed in Unity3D, but when you started, there was not a Unity 2D Toolset, right?

That is right, it was developed later. We researched the tools we needed and how it is easier for graphic artists to produce a 2D game efficiently. We developed these tools ourselves, for instance, a preview window, where you can see the game running. The changes on small details were therefore much easier to review. We developed some tools that made life much easier and we worked closely together with Unity. We exchanged some knowledge with





TOM HANKS & AUSTRIAN ANIMATIONS

Neuer Österreichischer Trickfilm

Interview by Lukas Kampichler

The new film “A Hologram for the King”, starring Tom Hanks, has been embellished with animations from Austria. Benjamin Swiczinsky of the studio Neuer Österreichischer Trickfilm (New Austrian Animation) has told us the details.

How did you get into this project?

The director Tom Tykwer had received a DVD collection of the Film Academy Baden-Württemberg, which contained my film Heldenkanzler. Eventually, my phone rang and Mr. Tykwer asked if I wanted to do it and if I had time to come to Berlin for the next few days, where the lynchpin of the production was taking place. The pre-production started quite quickly. So actually, it was just like you know it from stories: The phone rings and Hollywood calls.

How was the workflow for a project of this size?

We had to produce about two and a half minutes of animation and we only had a time frame from late summer to the end of the year. But it was manageable and clear to us that we wanted to do it anyway. For such a big production, we wanted to show our best. We also had some side projects and, being three partners at Neuer Österreichischer Trickfilm, I managed the “A Hologram for the King” project.

We quickly put together an animation crew with Santiago Lopez and Sascha Vernik, who were our two main animators. In addition, Guillermo Segui has animated and myself, of course. Hilde Renner, a colleague from the office, helped me with the clean up and colouring. Lukas Kampichler took over the compositing in the end. There was no help brought in from abro-

“ ...
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want to
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low-wage
countries
... ”

ad, although we initially thought that we would need to outsource, because there are not very many experienced 2D animators in Vienna. We could have outsourced to foreign countries, because there was no allegation of funding or something, but we decided to do it all in Vienna.

So nothing had to be outsourced to China or other cheaper countries?

No, not at all. With such a spontaneous project it would not have worked out, anyway. Our general attitude is that we do not want to outsource to low-wage countries.

On the one hand, because it is socio-economically questionable and on the other hand, because it is not sure that you really save money, if you want to achieve a certain kind of quality. Sending everything back and forth too often can be more expensive, than producing in Europe from the start.

How was the process of the concept phase? Did the director and the producers have a clear vision in mind or did you have a lot of creative freedom?

Within the script, it was relatively clearly defined what they wanted. We also added some ideas, together with Tom Tykwer and Ulrich Hanisch, his production designer, and finalized them. For this purpose, we met a few times in Berlin and elaborated on those points. There was even a layout of the movie available,

created by the director and the cutter, it consisted of found footage and was a very rough guide. But they said that they would prefer us to fill the animation sequence with our own ideas. You can say it was about 50 percent ideas from Tom Tykwer, Ulrich Hanisch and further 50 percent our own. I started to make storyboards and moodboards, to show how the final film stills could look like. It was a process that went back and forth, taking up the most time within the production.

And the designs also came from you?

Precisely. Conceptually, I have done quite a lot, actually the whole storyboard and design of sets, characters, etc. Only for animation, clean-up and compositing, I added some people to the team. After pre-production was the time for it, because then I had a clear vision of the project and could easily pass over a nice work package to the others.

Which software did you use for the project?

It was a mixture. We did the 2D animation in Flash and TVPaint and the 3D space and compositing was in Fusion. The concept draw-ings were pen and paper, of course.

You are mainly an animation film studio. Nevertheless, you take on film projects involving compositing. Have you helped with the VFX shots?

It depends on how you define VFX, but things like tracking and so on we did a lot, yes. But we shared some shots with RiseFX, who did the main VFX work. Finally, we exported the shots in different layers for the finishing at Arri in Munich.

Thanks to the internet, it is no longer a problem to collaborate over longer distances. How was the collaboration with Germany?

The places are thankfully not too far apart geographically, although there were also American co-producers such as New Line Cinema and Warner Bros involved. But we only had direct contact with Berlin. In the beginning, I kept flying to Berlin in the mornings and back in the evenings of each day. It was necessary as well as helpful for the progress, because Tom Tykwer did not have much time. He was work-ing on two other projects and was not that much in Germany, therefore he had no time and nerves to use Skype.

Are there any deleted scenes? Why?

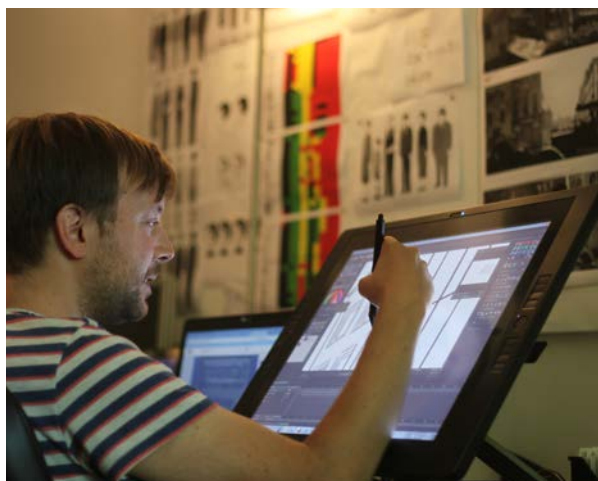
There were several test screenings, which resulted in scenes being deleted and changed. But which parts indeed did make it to the final film, we will just have to find out in the cinema. I cannot tell you more, because I do not know more.



Conrad Tambour
Benjamin Swiczinsky
Timur Tietze



Benjamin Swiczinsky
Conrad Tambour
Johannes Schiehl



Benjamin Swiczinsky

Are there any new projects? Can you already talk about them?

No new project have resulted from this one so far, although I am very excited about this, our very first international project, which will be screened worldwide. Our former projects can be watched online of course, but up to now they were restricted to German speaking countries. I can give you a short summary on some other projects we are currently working on: One is called "Auf Edits Spuren", which is based on a biographical novel by Peter Stephan Jungk, who will also direct the film. It tells the story of Edith Tudor-Hart, an Austrian photographer that migrated to London in the 30ies. She seemed to be a spy for the Russian and, somehow, also for the British secret service. It is a documentary with contemporary witness interviews and so on. No filmed footage of Edith Tudor-Hart does exist, therefore, we will tell the most important stages of her life in "Film Noir"-like styled animations, instead of trashy reenactments. A former "Asterix" animator will also be involved in this project, Manuel Galiana from Madrid, he did a lot and has somehow animated half of European animation history: Asterix, Lucky Luke and even Die Familie Feuerstein (The Flintstones). Santiago Lopez and I will also help to animate this. Another project with a former team member of the Hologram-crew, Sascha Vernik, will be "Fritzi war dabei" ("Fritzi was there"). It is a German-Austrian-Czech-French coproduction, still in a very early stage. The story is based on a children's book, which tells the story of a girl living in Eastern-Germany and her friend, who wants to get her dog back out of West-Ger-

many. The story shows the historical and political circumstances of that time in a child-friendly way - the fall of the Iron Curtain, demonstrations, etc. We are going to create the whole storyboard and animatic in Vienna, which is very great, because this way we can have a strong influence on the project.

Also under development is "Der Papierene" ("The paper man"). This is based on a graphic novel by Sascha Dreier, which tells about the Austrian Miracle team of the 30ies. Therefore, it is about soccer. The project is still in a very early stage.

We did finish some short animation shots for a Smart advertisement, and for the SWR TV-series "Ich kenne ein Tier" ("I know an animal"), we did produce one episode. It is a series for small children of which in every episode another animal is being introduced. For instance, the elephant or giraffe, while a narrator explains what is so great about them. We have done a mole episode for Studio "Filmbilder", which ended up being quite a short, charming, cute animated film.

Is there anything else that you want to mention?

Yes, five times a year we are organizing an animation meeting ("Animation Tricktsch"). Everyone is welcome to join. It creates an opportunity to talk about animation, anything else or just to have a beer with us in a friendly setting.



"Alpendorf"



"Fritzi war dabei"



"Maulwurf" ("Mole")



"Sisyphos"

WALT DISNEY

ANIMATION STUDIOS



DISNEY'S HYPERION RENDERER

Article by Simon Richter

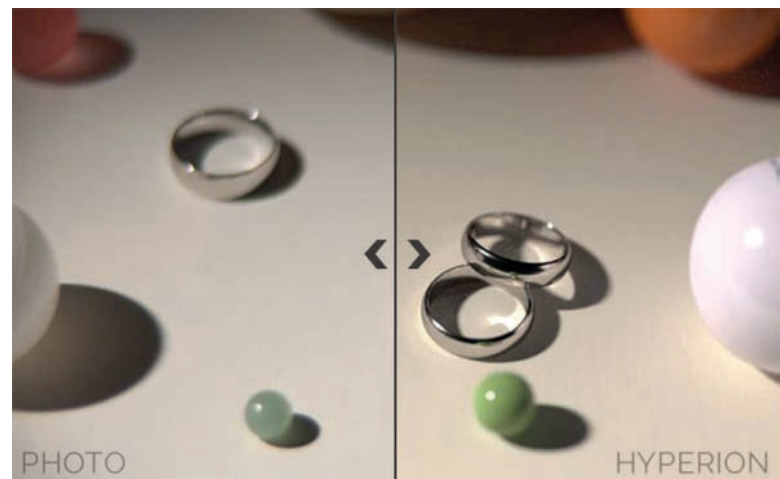
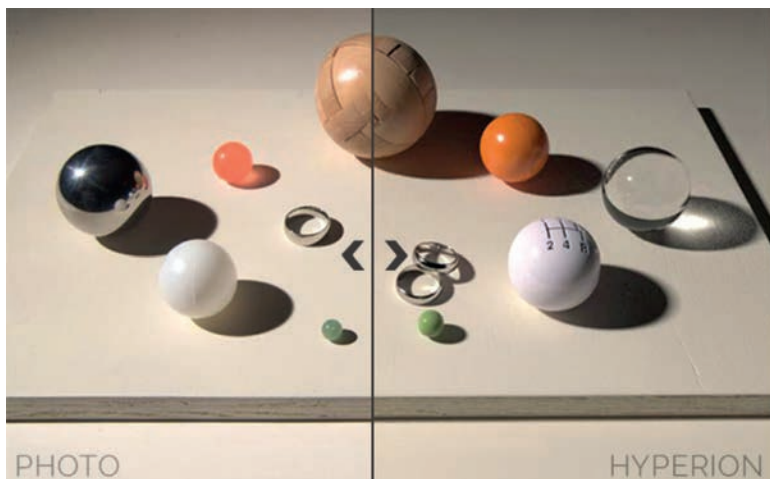
At this year's PIXELvienna Conference, Ralf Habel from Disney's Hyperion renderer team showed us insights into the making of the incredible movie "Big Hero 6" (German title: "Baymax - Riesiges Robowabohu").

Ralf started working on render technologies in 2005 at TU Wien at the Institute of Computer Graphics and Algorithms. In 2012, he started working as a post-doctoral researcher at Disney Research and two years later, he switched to Disney and started working on the physical based renderer Hyperion. **A lot of people in Austria know him, because he had been on the PIXELvienna organizational team from 2007 to 2012 and still is a regular attendee of the convention.**

Hyperion is the new in-house renderer for Disney's animated movies and makes things possible that would take ages to be calculated with other render engines. San Fransokyo - the city in Big Hero 6 - is a mix of San Francisco

" ...
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with other
render engines
... "

and Tokyo and it contains about 83.000 buildings, 26.0000 trees and 200.000 lights - all built procedurally. The whole scene has 340 million different objects, which is so much, that most current renderers would not be able to handle all of it. In 2012, Disney started the development of a new renderer with a new kind of path tracer for the upcoming new movie. By using a brand new render-engine the risk of failing or getting delayed was too big, so they used Pixar's Renderman as a backup plan. But, as the production went on, it took more and more effort to create the scenes for both renderers. So in 2014, the Renderman backup was shut down and Disney went all in for Hyperion. A standard path tracing technique bounces light around randomly, so there is no specific order



Real photographs of objects with very different characteristics, compared to re-creations rendered in Hyperion. (<http://www.disneyanimation.com/technology/innovations/hyperion>)

in which the scene objects are hit. For extremely complex scenes like in Big Hero 6, this can produce a high amount of non-relevant rays, which are waste of calculation time. Hyperion uses it's own specialized type of path tracing to reduce this problem. It bundles light paths, that are heading in the same direction, to be able to handle millions of light rays at the same time. So it gets more likely that rays in the same bundle hit the same object. And with this simplification, the renderer can optimize the calculations for the object that has been hit.

One of the main problems with render techniques like this is that these pictures contain a higher amount of noise even after several hours of render-time. The De-Noiser in Hyperion uses a special method that combines the different render passes, like not only diffuse, but also normal or depth data. Hyperion can do many things better compared to other renderers. Thousands of textures, giga-bytes of geometry, volume data and all globally illuminated by more than hundreds of thousands of lights is not an easy task, but with the scalability and the advanced techniques of sorting and optimizing, it was possible to render the whole town of San Fransokyo in one scene from far distance to street-level size, without using tricks.

But, even with Hyperion, the scene took about 86h/frame to render. To keep the production running, Disney increased its render power by adding four render farms with nearly 55.000 cores. Each computer comes with a dual 6-core Intel Xeon CPU and 256 GB memory. Capable of rendering 1.1 million render hours per day of the 15 GB geometry, 30 GB texture and 40 GB volume data. The movie needed over 200 million hours render time - more than Wreck-It Ralph and Frozen combined.



" ...
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Disney's Hyperion is not a photorealistic renderer and it is not planned to release a version for personal or commercial use. But it fits the requirements of a movie like Big Hero 6 perfectly and will be further developed to do the same for new challenges in the next Disney movie Zootopia.



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The ships of the participants
and of the spectators at the
"Antarian Transstellar Rally"

THE VFX OF STAR TREK VOYAGER BY LEE STRINGER

Article by Bernhard Raml

The title promises to stir the blood of every Trekkie and well, it kept what it promised. Lee Stringer gave us a detailed glimpse into the production of the Star Trek Voyager series, riddled with funny remarks and stories. However, the underlying theme of the talk was one of the key changes within the industry: The shift from miniatures to CG models.

He has been in the industry for quite a while now, working on titles like Firefly, Battlestar Galactica, Iron Sky and many, many more, but this talk focussed on the work he did for the Voyager series. Over the course of his career, he witnessed first hand the transition from miniatures to CG effects in the industry, and the Voyager series stands somewhat exemplary for this transition. They still used a miniature of the Voyager built by Tony Meininger, but also had a CG counterfeit, which was used for certain scenes. The problem with miniatures is that they are very costly to film and you cannot, for instance, crash one on the surface of a planet. But they are still much more detailed than the CG models. In turn, the richness of detail of the miniature helped the creation of the CG model, as it served as a great source and Lee did not miss the chance to point out the irony that this great advantage of the miniature was also somehow its own demise. Speaking

" ... a mix of
CG models
and miniatures
is the best way
to go ... "

with him later on, he told me that there seems to be a shift back in the industry towards using miniatures, as they still provide greater detail. In his opinion, a mix of CG models and miniatures is the best way to go, because even nowadays pure CG movies do not look as good as those who use a mixed approach. Look, for instance, at The Hobbit compared to The Lord of the Rings trilogy.

Another important lesson, which was taught by one of Lee's stories, is that you can achieve





Star Trek Voyager



“ ... you can achieve great things with smart usage of your resources at hand ... ”



Lee Stringer on Serenity Bridge

great things with smart usage of your resources at hand. One day, he was tasked with creating a close up, where the camera had to zoom from outside the ship to Captain Janeway through a window. The current CG version of the Voyager lacked the detail for such a shot, so he refurbished just the front of the ship, using the miniature as a reference. To make the scene more vivid he used some amazingly simple tricks: Only one character was rigged and he used some random motion capture to make him move around the room. To add some additional movement, he used blend shapes to blend between different poses of the stock characters. As the scene was quite small, the viewer would not notice that these were just some random movements, but it added to the realism of the scene - all made possible with little use of resources and some creative thinking.

All in all, the talk was one of the most entertaining talks at the Pixelvienna X and additionally provided a great wealth of experience, which is of tremendous value for anyone who wants to set foot into this industry.



BLIZZARD TALENT DAVID LUONG

Interview by Kevin Nilles

“ ... I recorded the audience reactions, so I could see how the people were cheering and laughing ... ”

How did you become an artist?

I took four years at Academy of Art in San Francisco, after that I focussed on visual effects and animation. I found that I love compositing and matte painting. I graduated and I wanted to become a compositor and matte painter. And, I love games and films so I hoped that I could work there. After graduating, I did a two week join test for Disney toon studios, but I discovered that they wanted someone with more experience. I searched for some more and got a job at Luma Pictures Studios in California, I worked on “Underworld: Evolution” for 6 months. After that, I worked for Rhythm And Hues Studios. There, I worked on “Garfield 2”, “Superman Returns”, and “Night at the Museum” as a compositor and also did matte painting stuff. A friend of mine, Jason Hill, also worked there, but he left early to work for Blizzard. He kept contact with me and then later asked me, if I would want to work for Blizzard. I was so happy about this. I sent my matte painting and compositing work to them. The next week, Blizzard gave me a phone call and I went out to an interview in California. A week after that, they said: “you are hired”. So in November 2006, I started at Blizzard and now it is my ninth year there.

I have seen that you worked on all the cinematic trailers of the World of Warcraft, so that is a long time...

All the cinematics from “StarCraft 2”, “World of Warcraft”, “Diablo 3”, “Overwatch”, “Hearthstone” and it is an honor to work on all of these cinematics.

How does it feel, seeing all these people being fascinated, when the cinematic trailers are launched?

We worked hard on the cinematics and sometimes, we think, “ooh it is a lot of work and much pain”. So, last year, on BlizzCon, we showed the cinematic from Overwatch and I recorded the audience reactions, so I could see how the people were cheering and laughing. The reactions were amazing.

What tools do you work with? More on analogue side or more with digital tools?

Most of my work is digital, with a tablet, and I integrate a lot of my photography into my matte paintings. Most of my textures are own photographs, I also took a lot of pictures in Vienna.

What are your future projects?

I just finished work for the cinematics of the game "StarCraft 2: Legacy Of The Void". The next will be the new "World of Warcraft" and "Overwatch" cinematics.

Do you have time for personal projects at Blizzard?

I do not have much time to work on personal projects, but I focus on my art gallery, it is downstairs where I live. And I talk a lot with other artists and exhibit their works on my walls.

Which tips do you have for people who also want to work at a company like Blizzard?

It is really hard, because you are competing with everyone from all over the world. And, they also hire only the best people in the world. So you have to try to get a lot of experience in the gaming or film industry, build up your portfolio and practice every day. If you want to, for example, work in modelling, then model every day. If you want to be a matte painter, you should train your 2d and 3d skills. And continue photographing, because you can use a lot of your photos for matte painting.

Practice is important. But it seems also to be important to stay in contact with a lot of people...

Yeah, keeping contacts with tools like Facebook, Instagram, LinkedIn and CG websites is good and important.

How do you like PIXELvienna?

I am so happy, that I can be a speaker at PIXELvienna. I am at BlizzCon next week and have deadlines, but I had a little spare time, so I took the time to participate.

" ... build up your portfolio and practice ... "



Digital Matte Paintings "StarCraft II: Legacy of the Void"
(http://www.davidluong.net/wp-content/uploads/2016/01/p070_all-three-watermarked.jpg)

Digital Matte Painting created for Ballistic Media Publishing's Book
"D'Artiste: Matte Painting Vol. 3"





iPhone5 "Dogfight"
Screenshot of the game
"Ace Ferrara & The Dino Menace"

ONE MAN, TWO YEARS & ACE FERRARA

Interview by Alexander Grenus

Philipp Seifried, the sole developer of Ace Ferrara & The Dino Menace, gave an interesting talk about visual effects at PIXELvienna X. Since I am a huge fan of the game (and dare I say a friend of Philipp), I jumped at the chance to interro... um... interview him.

Please introduce yourself to our readers...

Hi, I am Philipp. I started working on games around ten years ago when I wrote the code for the prototype of an Austrian online world, that was called Papermint and I have been interested in games ever since. I released a small Flash game called "Drifts" around 2005, that had a lot of players and later on I had the idea to co-found a startup in the Flash and advertisement games fields. I worked two years on the mobile game Ace Ferrara & The Dino Menace. In the past year I worked as a programmer and, to some degree, also as a game designer at the Vienna based company Socialspiel.

When did you start to develop interest for game development?

I had my first computer when I was six or seven, it was an old Commodore 64 and I started typ-ing in all the program listings that I could find in magazines.

I programmed my first little game at around the age of eleven in 20 lines of Basic. It was a car chasing simulator for a contest of a magazine.

" ...
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something like
that myself was
something that
drove me
... "

For a while, I was not that interested in games. From 2001 to 2005, I studied information design and during that time, I was interested in many different things like motion graphics, 3d animation and media art.

Then, after studying, I looked for a field where I could apply all my different kinds of interests. Games were sort of ideal for that, so that is how I got into that field.

When did you get the first idea for Ace Ferrara?

That was quite a long time ago, when we started with our Flash game startup "Bobbiebrook" back in early 2008.

I already had a prototype of a Wing-Commander-Style-Game with a custom made 3d engine in Flash, and you could already dogfight in there.

For a while, our two man startup worked on a project, which was very similar to Ace Ferrara. But it was an on and off project, that never really got anywhere. So, after we closed the com-

pany, I really had a lot of ideas that were still floating around and I was trying to get that particular project off the ground.

“Wing Commander” was my first big game that I played for hundreds of hours back in 1991 and 1992, and the idea of making something like that myself was something that drove me and really felt great.

In Ace Ferrara, you have a lot of - let us say - crazy characters in there. Where did you get the inspiration for those?
You mentioned in your talk that the game is inspired by 80s cartoons. Which specific details inspired you?

I think there is a lot of stuff floating around. A lot of those are based on pretty general tropes, like in a lot of those cartoons, you have as an annoying kid.

For instance in M.A.S.K. you had this guy who was the son of the chief and he had a little robot. In the original “Battlestar Galactica” series, you had a kid with a robotic bear or something, and in “Inspector Gadget” you had that smart little girl with her dog.

So this sort of kid with animal or robotic sidekick trope is very prevalent, I think. For most of the characters it is just stuff that is really floating around. I also picked the characters for their identifiability, like you would have that scientific genius, that is an old trope. You would have the cybernetic cat, that is arrogant and the top pilot, I think that is also a bit of a cliché and playing with it, was how the cast pretty much got together.

“ ...
 I also
 picked the
 characters
 for their
 identifiability
 “ ...

You mentioned the cat, you have cats, did you get any inspiration from them?

The cat characters are older than my cats actually are, in the sense that the concept for the characters was around even before I started working on the project alone.

But, at that point, it was just an idea, and the personalities, yeah, the cats definitely inspired those.

The icon of Ace Ferrara and the profile pictures you use, are quite similar. Are you somehow Ace Ferrara?

No, not at all. I mean, I think you always draw from experience when you are writing characters. One thing that defines this guy is that he is very young and he comes into a very elite group, where he is somewhat in above his head and he is a bit of a “know-it-all”. That creates some conflicts and awkward situations. For that, of course, there are some experiences I could draw from, but I do not see a lot of similarities. I think these are aspects of my own experiences that went into him, but I did not model him after myself. Not consciously, anyway.

How long did you work on Ace Ferrara?

Two years in total and at the point of the Android release, which was about a month after the iOS, I had spent almost exactly 3500 hours on it.

Do you still work on it from time to time?

No, I cannot, I do not really have the time at this point. I actually wish I could, but it is one thing to work on something like that full time, when

Game-Developer Philipp Seifried





iPad Retina Screenshot "Shuttle"



iPad Retina Screenshot "Cap Ships"



iPad Retina Screenshot "Dino Dialogue"

"... it would have helped, if the gameplay itself deviated more from its inspiration ..."

you can work all day and then continue in the evening. It is another thing, if you have a full-time job. Even if you take every weekend and spend 8 hours a day on Saturday and Sunday on the game, that amounts to around 70 hours per month. Given the total scope of the project, that is next to nothing, so it would take a really long time to further develop the game in any meaningful manner.

What was your goal for Ace Ferrara?

The main goal was always to create a commercially viable game, not artistic self-expression.

But within that constraint, I wanted to make something charming, over-the-top and funny. Most space combat games have a very militaristic, bro-in-space sort of vibe that I find a bit boring. I thought there might be a niche in combining the genre with a light-hearted story built on tropes from 80s cartoons. The game was not supposed to be a parody of those type of cartoons, though - more like a warm nod to them.

You have got a lot of good reviews for the game, but sadly it did not take off.

When did you realize that it was not a commercial success?

The thing is, that the press by itself does not really sell games, not if you do not have ten thousand articles out about your game. To be honest, the first time I thought it would not work commercially, was almost a year before release.

Because, it started out, that Free2Play games were becoming a big deal, but there were still niches and a lot more success stories of unknown developers working on paid games.

Around 2013, I think, that really died out. So I guess that at the beginning of 2014 - and I released in late 2014 - I already felt, "well this would probably need to be a very different game in order to become commercially successful." But, by that time, I had already spent a year on it.

There is always the chance to get a huge feature and big media picks it up, but I did not expect that to happen. I figured out, I would give the game about a month or so until I would really admit to myself, that this was not commercially successful.

If you would know what the outcome is, would you still do it again?

If I have not done it before, yeah, I think so. I would probably pick something more artistic.

I mean, it was a project I loved very much, and I put a lot of heart into it but it was still a project designed with commercial constraints in mind. If I had to do it again, and I would do it again, and I already knew that indie games in 2015 are a very risky proposition, then, I would probably go all in and do something that I think would be artistically more daring, but, not necessarily full fledged Arthouse type stuff.

Ace Ferrara was intended to be a commercial product and I tried to put in a unique and fun angle, story and all that stuff, but it is not big art. If I had another two years to do something like that, I would probably take it into a direction even more self-expressive.

You put a lot of love into the game, however, what parts of the development process did you enjoy the most?

There are a lot of things I really enjoyed doing. One thing that stands out is, when you put together the AI for a game - on the programming side - you make all those clever notes. Then, you code it up and it is an awesome moment when you have opponents in there, that show interesting, fun and non trivial behaviors for the first time and react to your actions, that is really great.

Another thing was the title track, which I am quite proud of. I am by no means a professional musician, but I think, it really captured the spirit of those TV-shows that inspired Ace Ferrara. It is awesome to sit down with a guitar, a bass and a keyboard and have something like that come out.

From the work process I would say, that creating all the 3d models, textures and stuff was really awesome as well. Because, I am not as much of an artist as I would like to be, but I had a lot of great evenings working on a model on one screen and then watching a whole season of some TV show on another screen, that was a great way of working.

Do you want to share any lessons you learned during the project with us?

I think making room for more changes during the project, would have been a good idea. I set out to do a mobile version of Wing Commander with a light-hearted story.

I think I made a lot of game design decisions, that made sense for a mobile platform, like reducing the gameplay to 2D. But, I think making the game look more remarkable for the first 20 minutes, or when you pitch it to a journalist, could have gone a long way. I think that the

game has a bit of a problem starting with a long tutorial. It really needs a bit of trust from the player in order to build up to the games potential and to get the player their payoff.

So I think, I lose a lot of players at that point and I do not have a mission in there, that is specifically tailored to just showing it around to journalists or at events. I think that it would have helped, if the gameplay itself deviated more from its inspiration and matched more the wacky storyline. I am thinking about big huge weapons, flamethrowers, knives at the edge of your spaceship and stuff like that.

Just stuff that you show someone for the first time and they are like "Woah, that is way over the top! I want to play that."

If I had allowed it to deviate more from the typical genre games, in order to impress YouTubers, first time players or people unfamiliar with the genre, that would have probably been a good idea.

We already talked about it nearly a year ago, however, do you still think about making a PC version?

I think about it every once in awhile. There are two issues here: On the one hand, if I go the PC route then I think I should fix the things, that I think did not work that well on mobile, like I should redesign the weapon system and adapt the humor of the gameplay a little bit to that of the story. And then, I should probably do other things that work better on a PC platform as well, like lengthen the missions and all that. So that amounts to quite a lot of work in total. Unless I really just do a very quick re-export on PC and put that on Steam.

But then, the more I change, the more I think it would probably hurt the game, because there is already so much information about the mobile version out, that will be then inaccurate and the two versions would be very divergent.

I am still thinking about ways to reconcile those things. It would need a slightly different title, just to make sure it is a different game and then that would probably mean maybe doing something with the story as well, just to give mobile players some different aspects to play it.

It is hard to estimate what of all the things, that I want to change would need a change. So I am still putting it off for a little while, but the mobile game has been out for a year, so it is not an urgent thing, I do not profit from the mobile game anymore.

So you would really like to work on Ace Ferrara again, but you currently do not have the time?

I would have happily continued for another year working on expansion packs or a sequel or something. I mean it is always the question, how much time do you put in something like that and what are your chances of success. To have success on Steam is not easy anymore. Coming out of that and having to leave that little family in the game behind was hard for me, so I would very much like to continue doing something with it.

"... how much time do you put in something like that and what are your chances of success ..."



Game-Developer Philipp Seifried

Caty Cosmos & Judy Jupiter
of Fijuka Space Patrol

by Anselm Hartmann
& Marie-Thérèse Zumtobel

FILM ACADEMY VIENNA PANEL: HOW MUCH VFX IS ENOUGH IN AUSTRIA?

Summary by Katja Jäger & Martina Fröschl

The Film Academy Vienna was founded in 1952 and is therefore a traditional education centre for filmmakers. International success and a ranking among the top film schools of the world do honor the institution of the University of Music and Performing Arts.

The Film Academy Vienna has recognized the signs of time and has established their own masters degree in Digital Art - Compositing. Therefore, it is interesting to see, what students can expect of this course at Film Academy Vienna. That was one reason to organize a screening of visual-effects-heavy works at this year's PIXELvienna Animation Festival. There were two music videos and a short film shown, followed by a question and answer session with the filmmakers. In addition, there was the possibility to talk to students of the Film Academy in the exhibition area of PIXELvienna. Not to forget the panel discussion about "How much is enough VFX?". The discussion was moderated by course director Martin Stegmayer and course assistant Valentin Struklec. The people on the panel were: Dominik Hartel, Thomas Kiennast, Severin Fiala and Veronika Franz.

As an introduction, the audience could watch trailers of recent well-known film projects of the directors at the panel. Therefore, trailers of the movies "Beautiful Girl" and "Attack of the Lederhosenzombies" from Dominik Hartel, "Ich seh, ich seh" ("Goodnight Mommy") from Severin Fiala and "Das finstere Tal" ("Dark Val-

" ...
I involve
the VFX people
during early
stages of
production
and we try to find
in-budget
solutions
... "

ley") from Thomas Kiennast were shown. After the introduction, the discussion started with the question:

Which VFX shots in your films were already planned in the script?

Fiala: Some points were already planned in the script of "Goodnight Mommy", for example the cockroaches crawling out of the flesh. We tried different approaches with greenscreen, steady-cam, motion control,... but other VFX shots just came up during the production.

Hartel: I keep the effects in my films as practical as possible, but there are certain shots, where I think VFX shots are very necessary. If that is the case, I involve the VFX people during early stages of production and we try to find in-budget solutions. We do no previz, but involve the experts in the discussion before, sadly, sometimes I even have to cancel VFX shots, when I think it gets too complicated.

Is it hard to do VFX on budget and did pre-visualization help your productions with calculation?

Kiennast: I find the collaboration with VFX studios not hard, but I also have to find ways to do the VFX on budget. I did some previz, but I think that a lot of the process comes spontaneously, especially in comedy, timing in the editing is important, so there sometimes have to be flexible budgets.

Franz: We get usually only small percentages of the budget funded for VFX.

How important are subsidies for your films?

Kiennast: Subsidies are important, especially for Austrian filmmakers. I think that is typical for our country and makes good arthouse films possible. Special subsidies for VFX are not available in Austria so far, but in Germany there is special funding.

Fiala: Up to now, you have to justify your decision to use VFX in your films. Funding agencies often do not understand the necessity for the story.

Kiennast: Nowadays, we have to cooperate very closely with the visual effects studios in early productions stages. It is not anymore only camera and lens for the creative process these days, there is so much more you have to consider for the allover visual style.

Is the increasing of resolution the right direction?

Kiennast: The technician in me says "yes", but the artist in me is a little concerned. You have to consider already so much for the visual decisions and style...

Where did you come up with VFX as visual decision in your projects? Did you have luxury VFX Shots or was everything "needed"?

Hartel: I had, for instance, a travel map like in Indiana Jones, that I used as a running gag throughout the film, these graphics were very important to me.

Kiennast: In compositing shots, for me, it is the best to combine practical and visual effects.

Fiala: We had to do the cockroaches, but I would have added more, if there would have been a bigger budget.

What do you think of the international situation and higher budget productions right now?

Kiennast: Big VFX movies and cutting so fast will be something for the museums soon... it is an overkill, a roller coaster...

What would you have done differently with a bigger VFX budget in your recent projects?

Hartel: It is a thing to think about the audiences. For "Lederhosenzombies", the audiences liked the practical effects with puppets.

3D-animation can be even with larger budgets bad - so it would depend on so many factors to solve something like Lederhosenzombies with a lot of CGI.

Fiala: Yes, I also think a bigger budget will lead to better visual representation of the story, the look will improve, but it is not necessary to add more VFX shots then. The story should be more important.

Franz: We were asked about our good-looking VFX shots at festivals - such a big compliment to Austrian VFX artists!

Hartel: With more budget you could tell the story "bigger" with more set-extensions and so on. The storytelling should be the most important, but larger scenery could improve the story.

Franz: Marketing is a big thing. We do not reach teenagers at the moment. There are no young distributors here - in the U.S. it is different, there is everything for that target group online...

Kiennast: For me it is a decision: Show an amusement park or tell a story. If we go and watch one of the old Alien parts - there was a lot happening hidden in the dark. If you look at the movies Jurassic World or Terminator 4, for instance, it seems more like a roller coaster ride that shows most of the action.

Hartel: There is a major difference between film makers and audience. Big budget movies are easier to understand for young audiences, because they are used to these kinds of productions.

Fiala: Yes, but that cannot change my approach - it is still my artistic decision. I do it my way.

Franz: Pupils do not go often enough to the cinemas - they do not have a film knowledge.

Kiennast: Yes, "Inside Out", for instance, was a great film, I would call it a master piece.

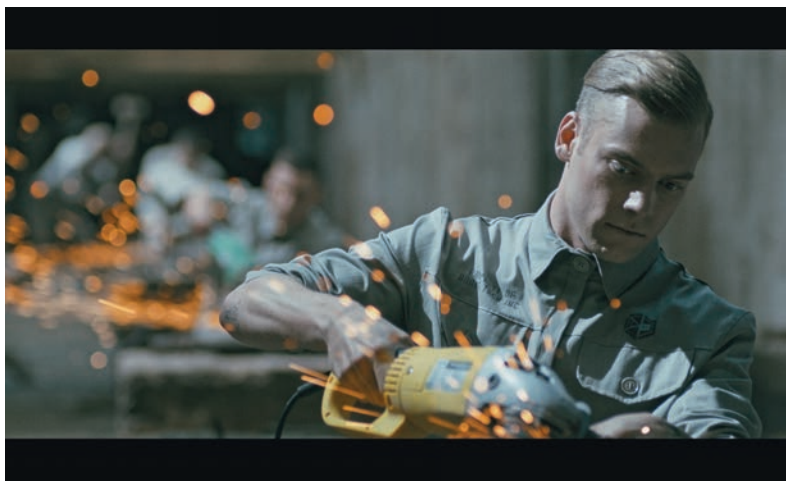
Hartel: In Austria, we have to face a fact: We have to be very creative with our budgets.

Kiennast: Yeah, at the moment, it is really not easy to show the effects that you want within the budgets given.

"... it is the best to combine practical and visual effects ..."

Music-Video
"Revolution"

by
Sebastian
Mayr &
Michael
Podogil



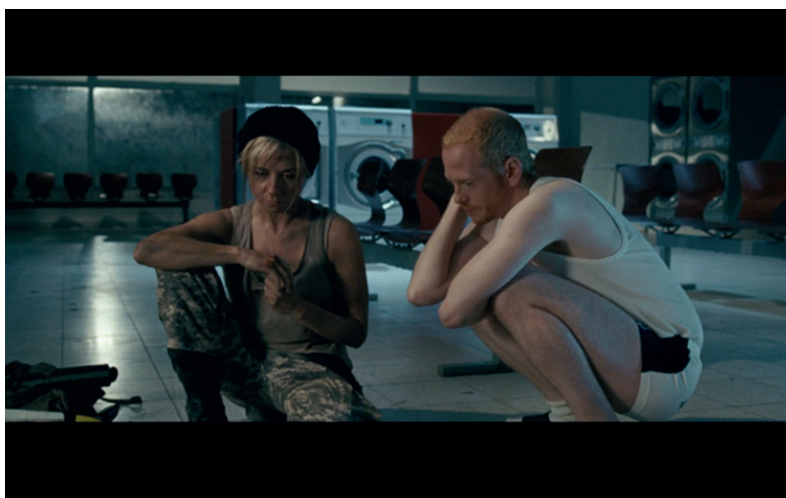
FAZIT:

Struklec: I think it is really great what has happened during the last couple of years. I see that VFX studio do expand. I thought we had a natural growth limit of around ten employees for each studio. Now we have studios in Vienna that are multiple times bigger than this. There were times during the last year where the demand for the work was so high, that we had to bring in people from other countries. I also noticed that there was a trend that artists would leave the country to find challenging work. Right now, it seems to me that a lot of artists have even decided to move back to Vienna to open up new studios. Still, international competition is, to be honest, on a complete different level and I have been thinking to myself: What is it that holds us back?

Stegmayer: We need larger post-production houses in Austria, because the artists have to work parallel and we also have to educate people. They should watch Austrian films. Larger VFX houses in Austria would bring back the huge amount of talented artists that have to work abroad. They would come, if there would be a project. There is a democratization process - the international industry is changing.

Shortfilm
"Tempus
Fug'it"

by
Béla Baptiste



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THE BEST OF THE BEST ANIMATION SHORTFILMS IN THE WORLD

PIXELVIENNA ANIMATION FESTIVAL 2015

Summary by Martina Fröschl

This year, the PIXELvienna Animation Festival got the great honor to open the conference. On the very first day of the conference, a diverse animation program was offered. For the very first time, there was a cooperation with the Film Academy Vienna. Three student projects with visual effects had been shown and the film makers where there to discuss their creations after the screenings with the audience.

The projects shown were a short film and two music videos: "Revolution" by Sebastian Mayr and Michael Podogil, "Ca Ca Caravan" by Anselm Hartmann and Marie-Thérèse Zumtobel and "Tempus Fug'it" by Béla Baptiste.

The response of the audience was great and there were a lot of interesting questions asked. The conversation with the young directors was moderated by Katja Jäger.

The second program focus of PIXELvienna Animation Festival 2015 was the current Computer Animation Festival selection of SIGGRAPH - so

to say the best of the world's best animated films. Thereafter, the program was rounded off by the latest Ars Electronica Animation Festival program introduced by Jürgen Hagler.

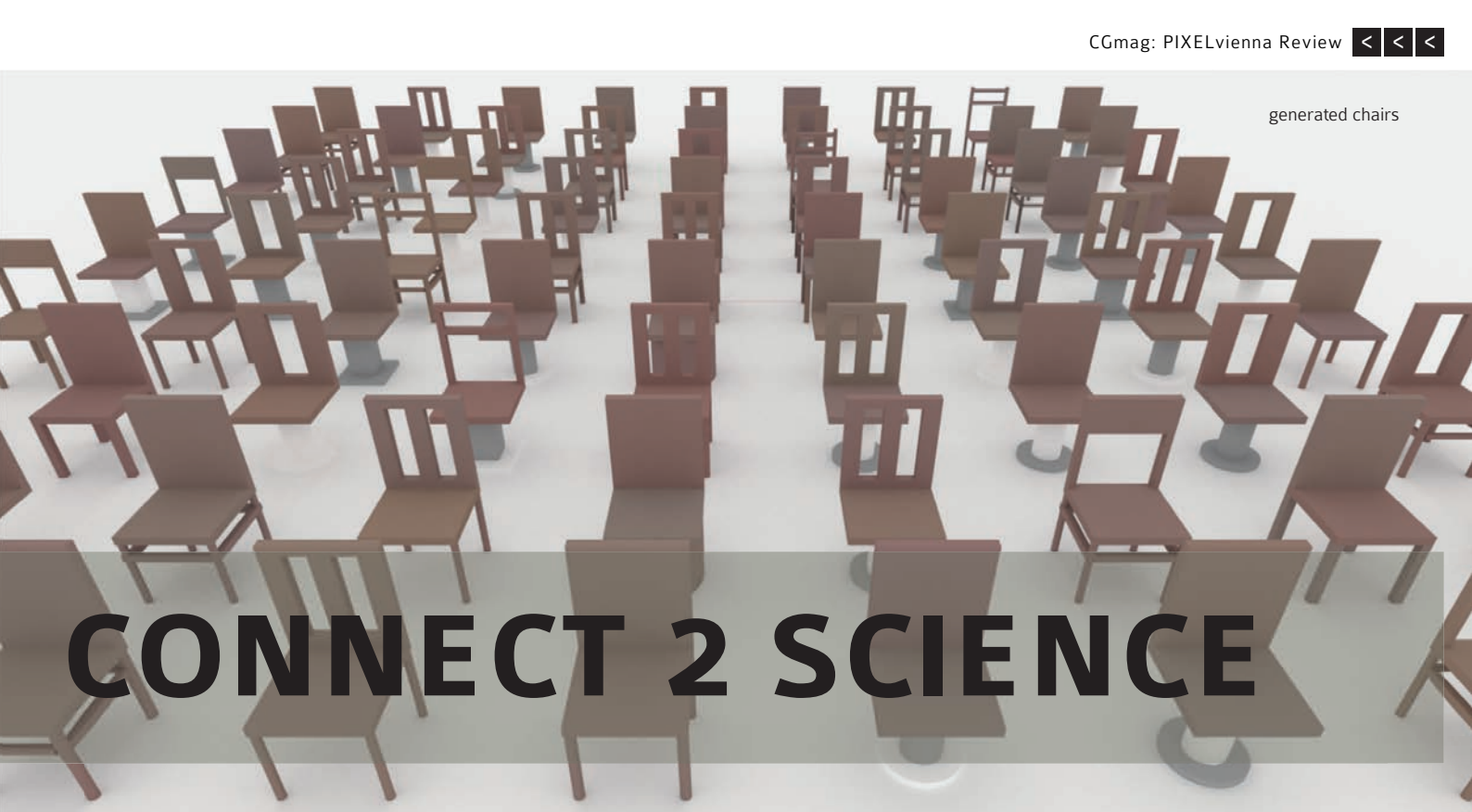
The cooperation between Ars Electronica and PIXELvienna has become a tradition. The conference team is proud of this cooperation, because Ars Electronica is nationally and internationally well renown for digital and new art.

For PIXELvienna X, the conference organization team had chosen the program cinema "Filmcasino" on Margaretenstraße for the screening of the PIXELvienna Animation Festival.

After a cinema day full of imaginative and inspiring short films, the attendees could relax and chat with drinks in the nice ambience of the charming cinema foyer. Before they went back out into reality, with the joyful anticipation of a PIXELvienna week-end full of amazing program highlights ahead.



generated chairs



CONNECT 2 SCIENCE

Summary by Martin Ilčík & Martina Fröschl

PIXELvienna is mainly a platform for computer graphics seen from the artistic point of view. Thus, as the technical coding, programming and research part is the essential base for every artistic image work with computers, the conference likes to provide also a presentation platform for technical research outcomes and work in progress in technical computer graphics. In 2015, Connect 2 Science the platform of PIXELvienna, invited again four researchers to present their current papers to the audience. The Connect 2 Science presentators were: Przemyslaw Musialski with his Consumer-Level Computational Design, Mohamed Radwan with "Smart Cutaways: Occlusion-Aware User-Guided Cuts on arbitrary 3D Shapes", Martin Ilčík with "Michelangelo: Cloud-Based Procedural Modeling" and Michael Grossauer with "Computer Animation in the Degree Program Multimedia Art". Here, we want to present the example of Martin Ilčík's "Michelangelo", because he is still searching for input from computer graphic artists and modelers.

Michelangelo - Procedural Modeling in the Cloud

This cloud service created by Martin Ilčík, provides the user with a convenient way of querying a vast amount of procedurally generated man made objects.

However, it is not just a simple key-word search, it also can take into account specific style request. To accomplish this, Michelangelo collects semantics for each object and its parts.

For instance a chair is not simply tagged as a chair, it also needs to have a seating surface and one to four legs. Such requirements are specified as goals for procedural refinement of the objects. Therefore, the appearance of legs is then described by even more specific goals.

The querying algorithm looks for procedural rules that fulfill the given goals. Each manual-

ly written rule-set can be automatically reused and recombined in many different ways and contexts.

Of course this reduces data duplication but also, given an extensive enough base library to draw from, would allow the user to create new objects and different expressions in no time.

The next step would be to create this base library and fill it with more complex objects. The cloud based nature of the project is very well suited to the task, as it allows for easy collaboration and every 3D modeller is encouraged to give it a try:

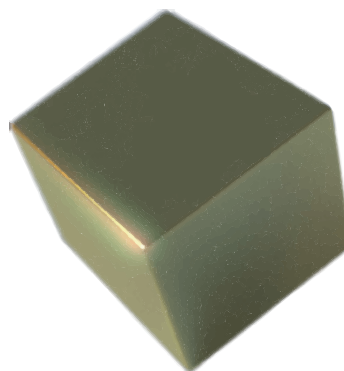
<https://www.michelangelo.graphics>

"... create new objects and different expressions in no time ..."

ACGA

AUSTRIAN COMPUTER GRAPHICS AWARD

Summary by Raphaela Burdis



Get to know the work of outstanding Austrian artists as well as the people behind the computer screens themselves at the Austrian Computer Graphics Award ceremony! Or participate and win the award and publicity for your mind blowing creation. The winners do not only get the fame and recognition that comes with the award – they are invited to give an insight in the making of their artwork via an interview that will be published in the Austrian CG Magazine www.cgmag.at. All pieces related to computer graphics can be submitted to compete.

The ACGA puts Austrian computer graphics into the spotlight and gives companies as well as individuals the chance to increase the public awareness of their work. In 2015 many outstanding artists handed in their work.

An international jury of CG professionals chose the best out of the nearly 100 entries, in eleven

categories. Once more it has been stunning how many sublime artworks were submitted by upcoming artists as well as professionals. It was absolutely great that some jury members like Ken Fountain, Ralf Habel and David Luong attended the ceremony and handed the ACGAs personally to the winners. Everyone could address them in a face to face conversation afterwards.

Take part and win the ACGA 2016!

Submissions for the 6th ACGA have started on February the 15th. Hand in your best work from the last two years or activate your brain power, use your genius imagination and create a magnificent project to participate in the competition!

Submit your work and find more information at www.acga.at

The ACGA Jury 2015:



Ken Fountain
Animator



David Luong
Senior Cinematic Artist,
Blizzard Entertainment



Rui Padinha
Senior VFX Artist,
MPC



Simone Grünewald
Head of Art,
Daedalic Entertainment



Juan Cañada
Head of Maxwell Render
Technology, Next Limit



Jarek Kolar
Senior Producer,
Hangar 13



Seungjin Woo
Concept Artist,
Behaviour Interactive



Ralf Habel
Senior Software Engineer @
Walt Disney Animation Studios



Thomas Liera
2D Artist, Sketcher for
Animation



Mario Ucci
Creative Director @
Passion Pictures



Andrei Riabovitchev
Concept Artist, Illustrator



Vassil Pepelyankov
CG Consultant & Supervisor,
CLX Europe



Bernd Diemer
Game Director,
Yager Development



Anselm von Seherr-Thoß
VFX Technical Director,
Incendii VFX



Christophe Hery
Rendering and Shading
Architect



Jose Lopez
Lead Character Designer,
Original Force



Colin Graham
Animation Director @ Ubisoft,
Montreal

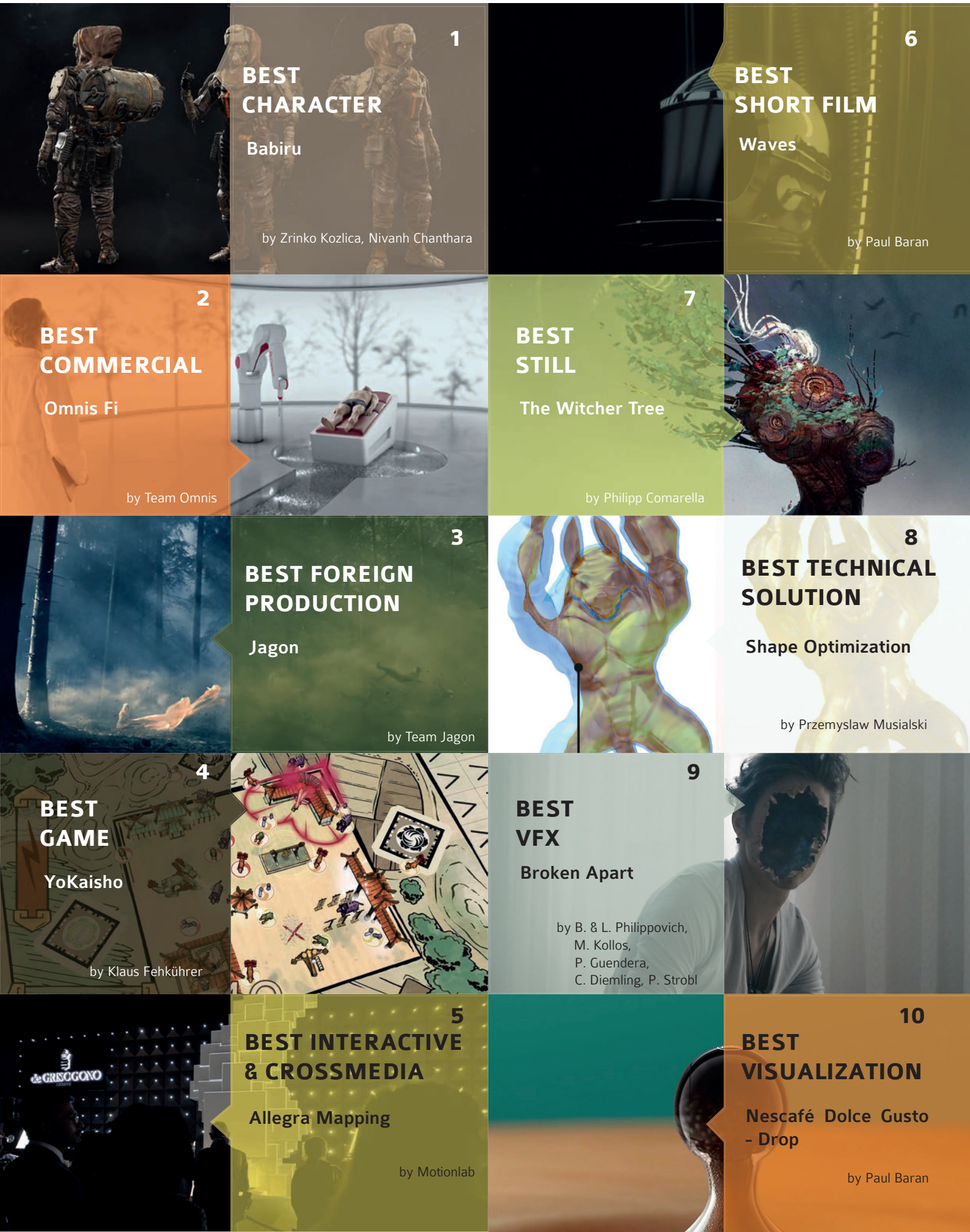


Michael Peter
Senior Texture and
Lookdev Artist, ILM



Benjamin Trobat
Storyboard and Concept Artist,
Illustrator

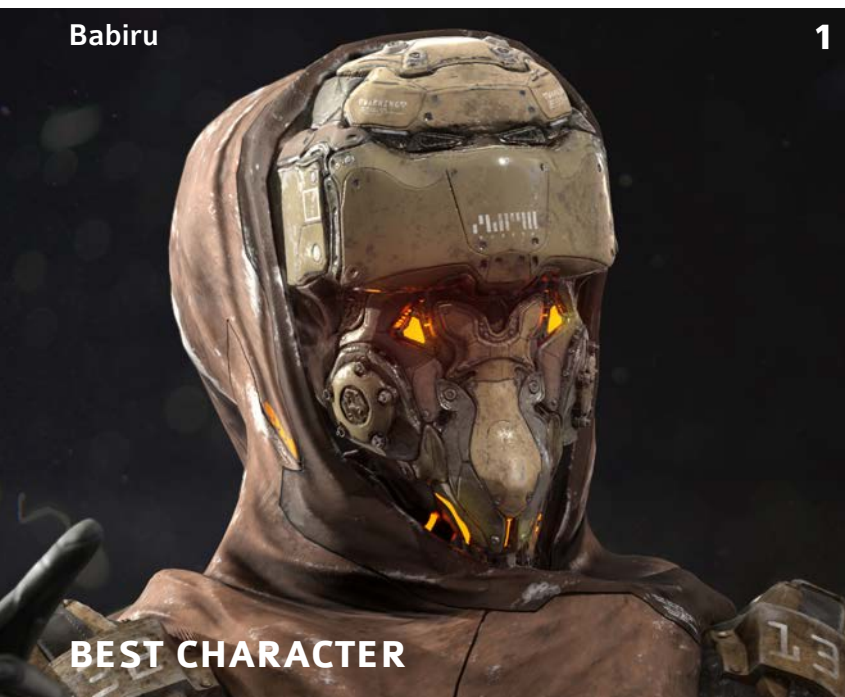
ACGA ARTWORKS



Babiru

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BEST CHARACTER



Allegra Mapping

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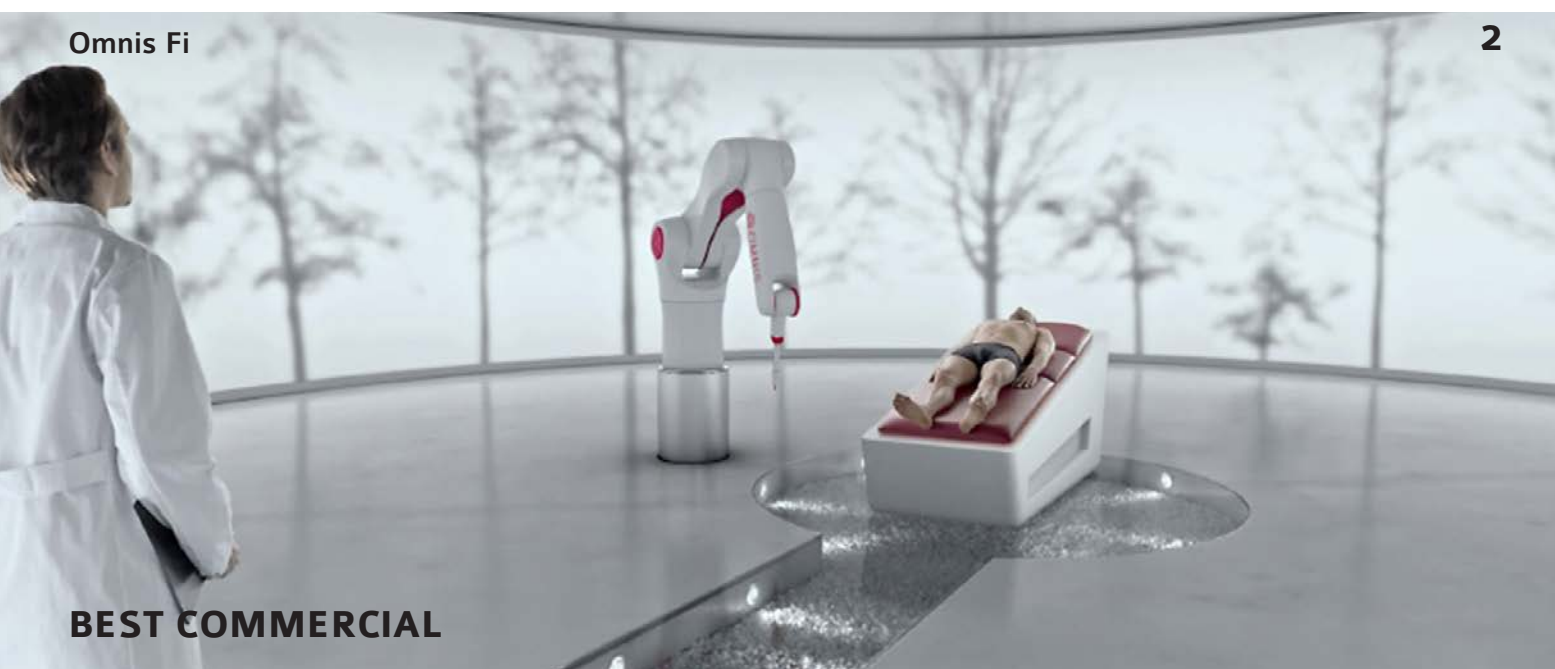
BEST INTERACTIVE & CROSSMEDIA



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BEST COMMERCIAL

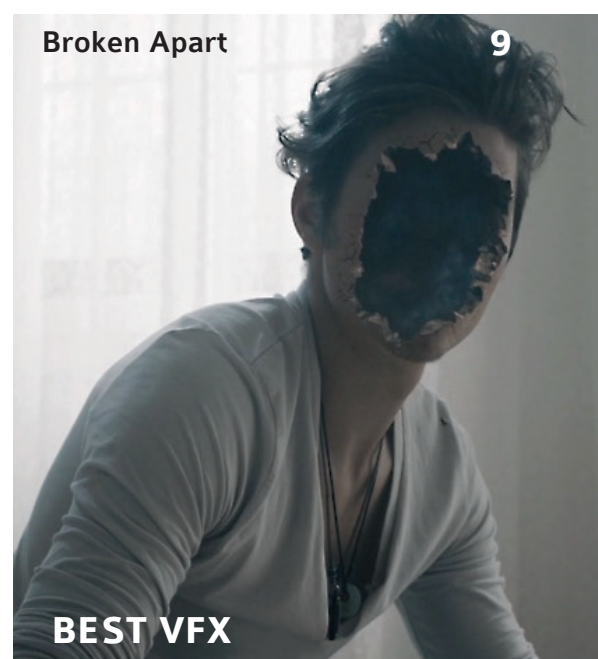


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BEST FOREIGN PRODUCTION







PIXELvienna Volunteers

Interview by Thomas Trautner

PIXELvienna is an annual conference on Computer Graphics and Animation in a form that is unique not only in Austria. First of all, because of the well-known speakers from all over the world and the remarkable conference sites, and secondly because of the fact that it is entirely organized by volunteers. Without their support, it would not be possible to organize such a successful and yearly growing conference.



What are the duties of a PIXELvienna Volunteer?

Generally we distinguish between two types of volunteers: First of all, there is the so-called "Core Team", which is a group of about 20 people that are planning the conference during the whole year. This work includes various aspects like contacting speakers, planning the PIXELvienna-Party & Battle and Animation Festival, technical solutions including ticketing and registration, social media and print, logistics and finances and of course organizing the volunteer program. That is why this team meets on a regular basis and discusses current plans.



Furthermore, there are the so-called "On-Site Volunteers" which work a couple of days before, during and after the PIXELvienna conference. They will be asked to perform a variety of tasks, for example taking pictures during the conference, bag stuffing, speaker and room assistance, coffee service, wardrobe and supporting the Animation Festival and PIXELvienna Party. Nevertheless, we try to take all wishes into consideration and make sure that volunteers have enough time to enjoy most of the talks even while they are on duty. As an additional thank-you to the full-access ticket, they will receive the well known PIXELvienna T-shirt, a unique souvenir which cannot be bought by attendees.



What else does the PIXELvienna Volunteer Program offer?

Distributed over the whole year, we organize regular meetings where volunteers can share experiences among each other. Additionally, we organize local studio tours like for example at arx anima, Friendly Fire, Industrial Motion Art, and many more. This will help volunteers to gain an insight into everyday work of people from the Computer Graphics industry. If they are lucky, they may even see some still secret projects and make up new contacts!

How can I become a PIXELvienna Volunteer?

If you want to be a part of either the core or on-site team, feel free to write to the PIXELvienna Volunteers Coordinator and I will be happy to welcome you into our PIXELvienna family!

If you do not know whether you want to support the core- or on-site team, do not hesitate to join a meeting and tell us what interests you. This makes it easier for us to find suitable and interesting tasks for you.

Already looking forward to hearing from you soon!

Your PIXELvienna Volunteers Coordinator



Thomas Trautner
volunteer@pixelvienna.com

PIXEL PARTY & BATTLE

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